

ACHYUT KULKARNI

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PROFESSIONAL EXPERIENCE

Data Scientist | Stealth Startup - IYKYK Inc. (Event Discovery Platform) | Charlotte, NC, USA May 2025 - July 2026

- Architected an event ingestion pipeline to power city-wide event discovery in Nashville, centralizing fragmented event listings across heterogeneous web sources and targeting 50% expansion of discoverable local event inventory.
- Drove implementation of Neo4j-based recommendation system that ranked events and users leveraging graph and location signals, enabling personalized event discovery and improving recommendation relevance by 28%.
- Engineered low-latency data pipelines with optimized query patterns and caching layers, reducing latency 40% under real-time load.
- Translated model outputs into product-facing ranking strategies, reducing churn 15% via improved discovery pathways.
- Implemented and deployed multi-stage ranking pipeline (candidate generation + re-ranking) optimizing precision-recall tradeoffs in production.

Graduate Researcher | Toyota Research Institute | Rochester, NY, USA Aug 2024 - May 2025

- Diagnosed false-braking failure modes in a self-driving perception pipeline using TCAV and CRAFT concept-level XAI; targeted retraining reduced false positives 31% and improved pedestrian detection accuracy 18% across pedestrian, cyclist and vehicle classes.
- Applied NMF concept extraction on YOLOv3 feature representations to surface safety-critical misclassification patterns.
- Cross-validated attribution findings with LIME/SHAP for model interpretability, producing multi-framework evidence for safety audit reporting.
- Contributed audit methodology & findings directly to AV safety playbooks, informing production-level safety validation & retraining protocols.

Software Engineer (AI/ML) | Megh Computing | Bengaluru, India July 2020 - Mar 2023

- Productionized real-time video analytics pipelines at scale across smart city, retail and finance domains using CPU, GPU and FPGA acceleration.
- Reduced edge inference latency 40% by optimizing YOLOv5/Faster R-CNN using TensorRT, OpenVINO and model pruning techniques.
- Operationalized cloud inference systems enabling reliable deployment in low-bandwidth environments, improving operational efficiency 30%.
- Optimized models against client-defined KPIs and latency SLAs, accelerating production deployment and reducing iteration cycles.

PROJECTS

Gemma4 Video Realism Audit | Gemma 4, Transformers, OpenCV, Multimodal Evaluation | Kaggle Gemma 4 Good Hackathon | [View on GitHub](#)

- Built a Gemma 4 multimodal auditing system for AI-generated video realism: extracted representative frames, detected synthesis artifacts, and ensembled repeated model judgments via consensus voting to rank synthetic outputs by visual convincingsness.

WebPage Builder AI | LLM Agents, Prompt Engineering, Ollama | [View on GitHub](#)

- Shipped a local agentic system that converts natural language prompts into validated, browser-ready web prototypes using constrained code generation, automated critique-and-repair loops and commit-gated validation eliminating manual iteration between prompt and result.

MedGemma Edge CT Triage Pipeline | MedGemma 4B, Hugging Face | Kaggle MedGemma Impact Challenge | [View on GitHub](#)

- Developed an end-to-end chest CT triage pipeline using MedGemma 4B with HU windowing, structured clinical prompts, controlled vocabulary normalization and scan-level urgency aggregation applying rigorous LLM evaluation methodology to critical diagnostic prioritization tasks.

TECHNICAL SKILLS

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, XGBoost, Gradient Boosting, CNNs, Transformers, Statistical Modeling, Regression, Classification, Clustering, NumPy, Pandas

Recommendation Systems & Graph ML: Neo4j, Graph Embeddings, Candidate Generation, Vector Search, Ranking Systems,

LLMs & GenAI: RAG, LoRA/PEFT Fine-tuning, LangChain, LangGraph, Prompt Engineering, Hugging Face, LLM Evaluation

Computer Vision: OpenCV, YOLOv5, Faster R-CNN, TensorRT, OpenVINO, Object Detection, CUDA, ONNX, Edge AI

Interpretability & Evaluation: TCAV, CRAFT, LIME, SHAP, NMF Concept Extraction, A/B Testing, Causal Inference, MMLU, TruthfulQA

Data Engineering: Airflow, Spark, PostgreSQL, MongoDB, Redis, BigQuery, ETL Pipeline Design, Data Modeling, Data Validation

Cloud & MLOps: AWS (SageMaker, S3, EC2), GCP (Vertex AI, BigQuery), Docker, Git, Linux, CI/CD, MLflow, Model Monitoring, Model Deployment

Programming Languages: Python, SQL, Cypher, C++, R

EDUCATION

M.S. Data Science | Rochester Institute of Technology, Rochester, NY Aug 2023 - May 2025

B.E. Information Science | BNM Institute of Technology, Bengaluru, India Aug 2016 - Aug 2020

RECOGNITION & CERTIFICATIONS

- **Winner - SCB Business Analytics Competition 2025** : Benchmarked GPT-4, Claude 3 and Mistral on MMLU, TruthfulQA and ARC; built governance rubrics for fairness, compliance and ROI; recommended Claude 3 as RIT's institutional AI platform. Ranked first. [Link](#)
- **Oracle Cloud Infrastructure AI Certifications (2025)** : Generative AI Professional, AI Foundations Associate, AI Vector Search Professional, AI Data Science Professional - validated through production RAG and vector search implementations.
- **Published Research, IJLTET 2020** : "Application to Detect Skin Cancer using CNN" - MobileNet, optimised for sub-\$50 edge deployment with explainable diagnostic outputs. [Read paper](#)